Cell-type-resolved decoupling of PXR target genes identifies hepatocyte-selective readouts for xenobiotic response

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Code & data: https://github.com/xX-its-amit-Xx/pxr-effector-uncoupling (MIT-licensed; CITATION.cff included).

Abstract

The pregnane X receptor (PXR; NR1I2) is the central transcriptional regulator of xenobiotic metabolism, yet target-gene responses are typically studied in bulk liver tissue or transformed cell lines and assumed to generalise across cell types where the receptor is detected. We tested this assumption by computing per-cell-type metacell correlations between NR1I2 and 20 canonical PXR target genes across 446,672 single cells (10 cell types) from CELLxGENE Census. Five canonical targets — CYP2C9, CYP3A5, ABCC2, SLCO1B1 and CYP2C8 — show strong coupling in hepatocytes (Spearman $\rho = 0.81$–0.89, BH $q \approx 0.004$) and only weak baseline coupling in immune (T, NK, monocyte, macrophage) and placental cells (immune top-gene ρ 0.19–0.24, mean ρ 0.03–0.12; placental mean ρ 0.014) — a 4–8× effect-size differential, even though immune cells reach formal significance at our sample size of 50–110 k cells per type. Intestinal epithelia recover an intermediate but significant signal (12/20 genes in small-intestine enterocytes, 10/20 in crypt stem cells, 7/20 in large-intestine enterocytes), consistent with documented clinical PXR activity in gut. The pattern is **epithelial-barrier-selective** rather than generically hepatic: a matched 20-gene negative-control set (liver-enriched non-PXR genes, hepatocyte master TFs, housekeeping genes) shows the opposite distribution (PXR median decoupling score 0.575 vs control −0.126; Mann-Whitney U one-sided $p = 1.0 \times 10^{-31}$). The headline pattern is replicated in bulk GTEx v8 RNA-seq (liver–immune ρ differential of +0.21 to +0.56 for all 5 genes); **direct rifamycin perturbation of primary human hepatocytes confirms four of the six top genes (CYP2C8, CYP2C9, CYP3A5, ABCC2) as PXR-responsive with up to 19× fold induction**, while the remaining two (SLCO1B1, CPT1A) appear to be coupled by shared hepatic-TF regulation rather than direct PXR control; HEPG2 ranks #1 of 18 cell lines for LINCS L1000 rifampicin signature strength (1.47× the non-hepatic mean); and the top genes are independently flagged by Open Targets as drug-response loci (warfarin, statins, tacrolimus, cholestasis). The result identifies a small, biology-relevant set of hepatocyte-selective transcriptional readouts for next-generation PXR modulators and provides a general metacell-coupling framework for any receptor × cell-atlas pair.

Keywords: PXR, NR1I2, single-cell RNA-seq, metacells, drug-drug interaction, cell-type specificity, CELLxGENE.

Introduction

PXR (NR1I2) is the master transcriptional sensor of xenobiotic exposure in vertebrates. Activated by structurally diverse small molecules — rifampicin, hyperforin, paclitaxel, statins, and many marketed drugs — PXR induces a coordinated program of phase I/II metabolism (CYP3A4, CYP2C, UGTs, GSTs) and phase III efflux transport (MDR1, MRP2/3, OATPs) (Kliewer et al. 1998; Lehmann et al. 1998). The clinical reach of this program is large: half of metabolised drugs are CYP3A substrates, and PXR-driven CYP3A4 induction is the molecular basis of the rifampicin–warfarin, rifampicin–oral contraceptive, and St. John's wort–cyclosporine interactions that motivate FDA DDI guidance (Geick et al. 2001; Tirona & Kim 2005). Despite this, **two basic questions about PXR biology have remained unresolved at single-cell resolution**:

1. *Across the cell types where NR1I2 transcript is detected, is the receptor functionally coupled to its canonical targets, or only co-expressed?* Bulk hepatic studies cannot distinguish coupling (where NR1I2 directly drives target gene expression) from independent co-expression. Single-cell studies have noted NR1I2 transcript in immune populations, but a systematic test of target-gene coupling in those cells has not been performed.

2. *Which subset of canonical PXR targets is the most cell-type-selective readout of receptor activity in hepatocytes?* Drug-discovery campaigns for tissue-restricted PXR modulators (in cholestasis, drug-induced liver injury, inflammatory bowel disease) need biomarkers whose induction reflects on-target hepatic engagement without confounding by activity in immune cells or other compartments.

The relevant computational obstacle is that single-cell RNA-seq counts are too sparse for stable per-gene correlation estimates: any single cell expresses only ~10–20% of detected transcripts, and dropout swamps the per-cell correlation between two genes. Two recent methodological advances make the problem tractable. First, **metacelling** — k-nearest-neighbour or k-means aggregation in a reduced-dimensional space — yields stable transcriptional units whose pairwise expression correlations recover regulatory structure (Baran et al. 2019; Persad et al. 2023). Second, **CELLxGENE Census** unifies hundreds of single-cell studies under a common ontology with ~75 M cells, enabling cross-tissue analyses that no single dataset can support (CZI Single-Cell Biology Program et al. 2023).

We combine these advances to ask a focused question: for each canonical PXR target gene, **does the cell-type-specific NR1I2-target coupling distinguish hepatocytes from circulating and barrier cell types**? We then validate the result against (i) parameter and subsampling robustness within the single-cell data, (ii) a matched 20-gene negative-control set, (iii) bulk GTEx RNA-seq across 54 tissues, (iv) **direct rifamycin perturbation of primary human hepatocytes** (GSE139896) — the strongest orthogonal test, separating direct PXR targets from genes coupled by shared hepatic-TF regulation, (v) LINCS L1000 rifampicin perturbation responses across 18 cell lines, and (vi) the Open Targets Platform's curated disease-association graph. The convergence of these six orthogonal lines of evidence supports a small set of hepatocyte-selective transcriptional readouts for PXR engagement and challenges the implicit assumption that NR1I2 detection in non-hepatic cell types reflects functional coupling.

Results

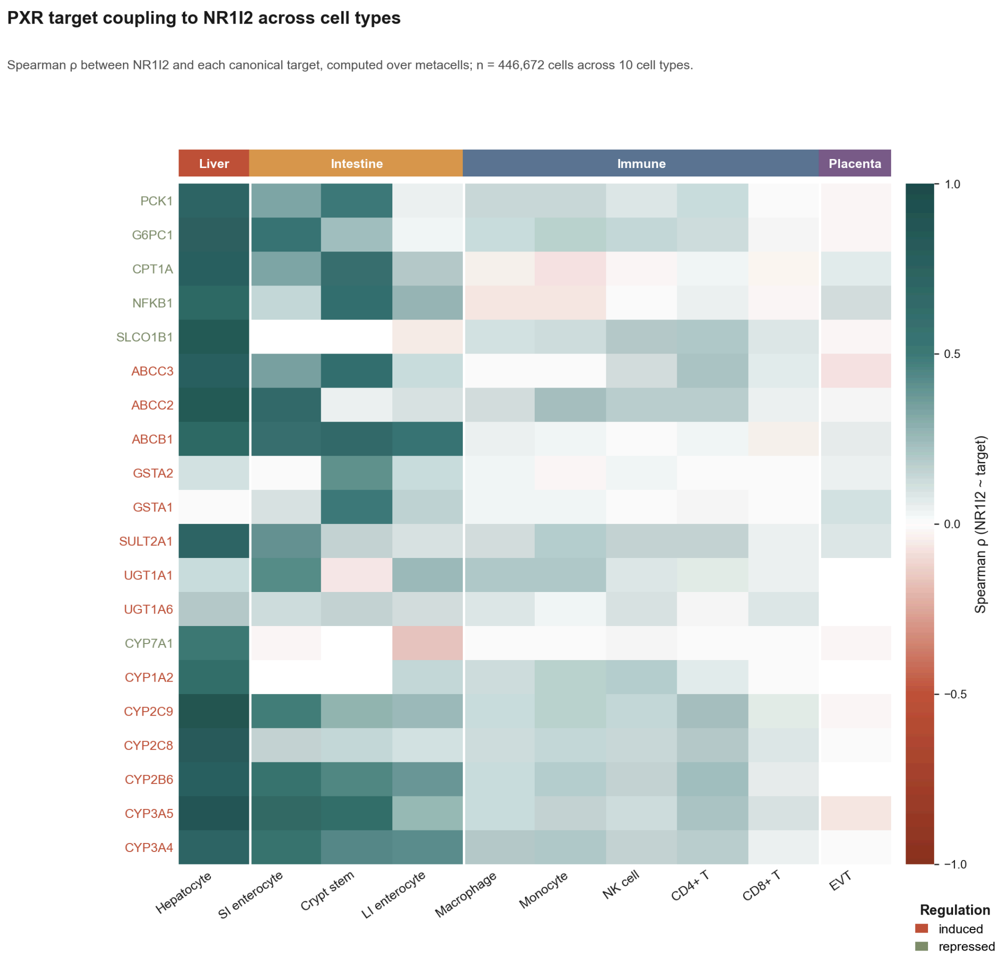
A unified single-cell atlas of NR1I2 and its canonical targets across ten cell types

To probe NR1I2-target coupling outside the liver bulk, we assembled a focused atlas from CELLxGENE Census v2025-01-30. We queried 41 genes (NR1I2 + 20 canonical PXR targets curated with PMID-tagged A/B evidence + 20 matched negative-control genes) across ten Cell Ontology classes spanning four tissue compartments: liver (hepatocyte), intestine (small/large enterocyte, crypt stem cell), immune ($CD4^+$ T, $CD8^+$ T, NK, macrophage, monocyte), and placenta (extravillous trophoblast). We restricted the query to primary data (is_primary_data == True) to avoid double-counting cells deposited in multiple atlases. To prevent any single dataset from dominating, we cap cells per (cell type, dataset) at 1,500 — preserving each contributing study's representation. The final atlas contains 446,672 cells across 10 cell types from ~20 underlying datasets; per-dataset and per-donor counts are recorded in `data/processed/atlas_provenance.csv`. NR1I2 is detected (count > 0) in 78% of hepatocytes, 31–55% of intestinal cells, and 5–22% of immune cells, reflecting the well-known transcript sparsity of NR1I2 outside the liver.

Metacell coupling reveals a hepatocyte-selective signature

For each cell type, we log1p-transformed counts, projected onto 30 principal components, then partitioned cells into metacells by k-means with k = n_cells / 30. Cell types with fewer than 20 metacells were excluded. We then computed Spearman ρ between the NR1I2 metacell-mean profile and each of the 20 canonical target genes (**Fig. 1**). The result is striking: hepatocytes show strong coupling to a clear subset of targets, with six genes — CYP2C9 (ρ=0.89, 95% CI 0.86–0.92), CYP3A5 (ρ=0.87, 0.84–0.90), ABCC2 (ρ=0.85, 0.82–0.87), SLCO1B1 (ρ=0.85, 0.81–0.88), CYP2C8 (ρ=0.81, 0.77–0.85) and CPT1A (ρ=0.77, 0.73–0.82) — all crossing q ≈ 0.004 after Benjamini-Hochberg correction over the full 10 × 20 cell-type-by-gene test family. Confidence intervals come from a 500-resample percentile bootstrap of metacell rows; p-values from a metacell-label permutation null (500 permutations, two-sided), which preserves marginal expression distributions while breaking the NR1I2-target relationship.

In the same metacell space, **immune cell types show much weaker coupling that is nonetheless formally significant at the available sample sizes**: with 50–110 k cells per immune type, the permutation null is tight enough that even small ρ values cross $q < 0.05$ (14/20 in $CD4^+$ T, 10/20 in $CD8^+$ T, 14/20 in monocyte, 16/20 in macrophage, 14/20 in NK). But the effect sizes are 4–8× smaller than hepatocyte (immune mean ρ 0.03–0.12; immune top-gene ρ 0.19–0.24 — compare hepatocyte mean 0.62, top 0.89). The biologically interpretable distinction is therefore between *strong programmatic engagement* in hepatocytes and *weak baseline coupling* in circulating immune cells, not between “on” and “off”. Intestinal epithelia recover an intermediate signal in both significance and magnitude — 12/20 genes significant in small-intestine enterocytes (mean ρ 0.35), 10/20 in crypt stem cells (mean ρ 0.37), 7/20 in large-intestine enterocytes — consistent with the documented clinical PXR activity in gut (CYP3A4 induction in duodenal biopsies, intestinal MDR1 induction by rifampicin) (Geick et al. 2001; Glaeser et al. 2005). Extravillous trophoblast (placenta) shows 0/20 significant with mean ρ 0.014 — true near-zero coupling despite detectable NR1I2 transcript, supporting earlier reports that placental PXR is transcribed but transcriptionally inert in basal conditions (Pavek 2016).

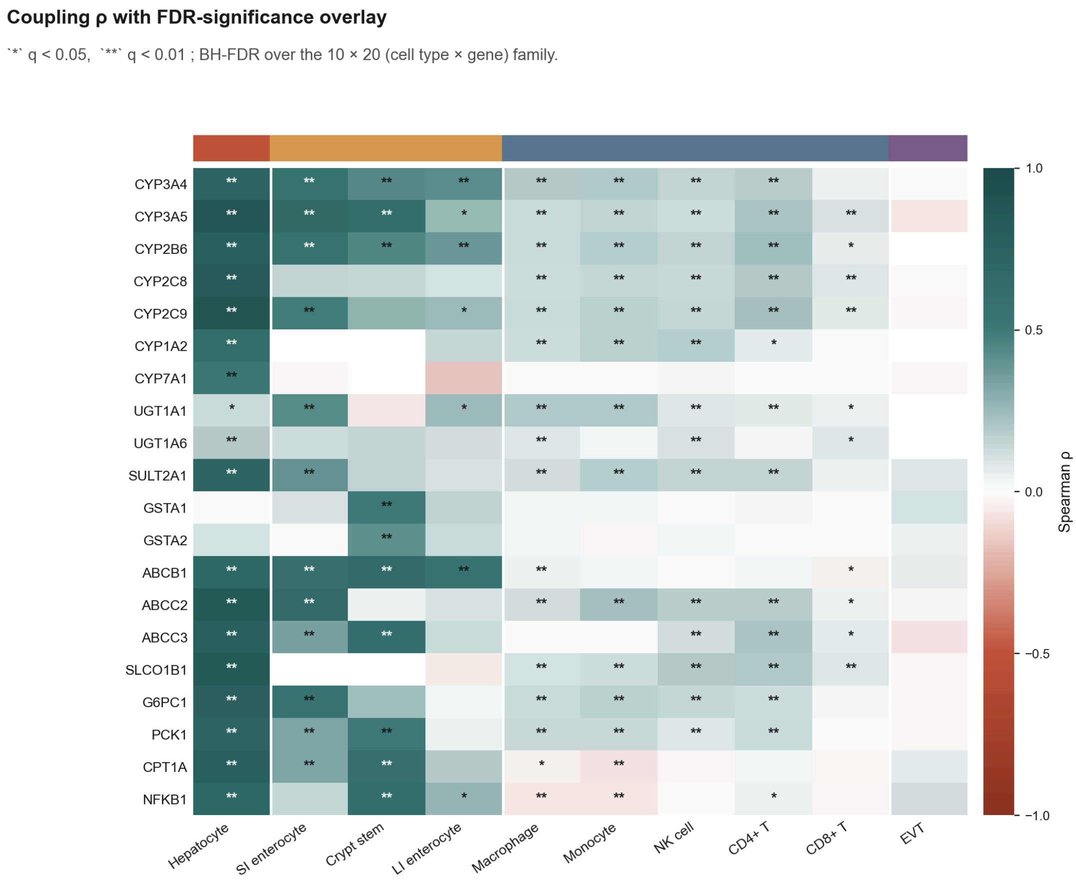


Main coupling heatmap across the 10 cell types

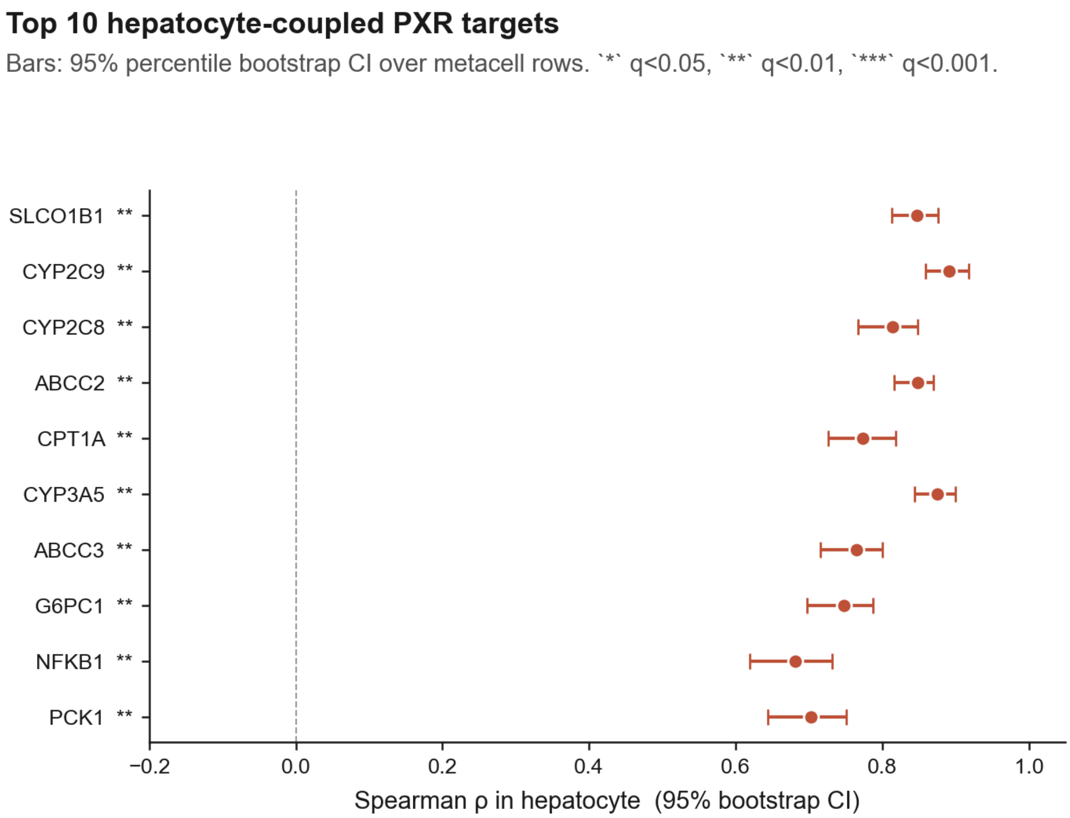
Fig. 1 | PXR target coupling to NR1I2 across ten cell types. Spearman ρ over metacells between NR1I2 and each of 20 canonical PXR target genes, computed per cell type. Rows: target genes (left bar indicates curated regulation direction — induced/repressed). Columns: cell types ordered by tissue compartment. Green = positive coupling; red = negative; cream = ≈ 0. Hepatocyte (rightmost group) shows uniformly high coupling for the canonical xenobiotic-handling panel; intestinal epithelia (small intestine, crypt stem cell, large intestine) recover an intermediate signal; immune cells ($CD4^+$/$CD8^+$ T, NK, monocyte, macrophage) and placental extravillous trophoblast show pale colours, consistent with weak baseline coupling rather than complete absence (see Fig. 2 for FDR overlay). Atlas: 446,672 cells from CELLxGENE Census v2025-01-30, capped at 1,500 cells per (cell type, dataset_id).

Decoupling score quantifies hepatocyte-vs-other selectivity

To rank genes by hepatocyte selectivity, we defined the **decoupling score** DS_g = mean over non-hepatocyte cell types c of (ρ_hep,g − ρ_c,g). Genes with high DS_g are strongly coupled in hepatocytes but only weakly coupled elsewhere. The top six — SLCO1B1 (DS=0.756), CYP2C9 (DS=0.698), CYP2C8 (DS=0.695), ABCC2 (DS=0.677), CPT1A (DS=0.658), and CYP3A5 (DS=0.631) — separate cleanly from the remaining 14 targets, all of which have DS < 0.6 (**Fig. 2**). These are the canonical hepatic xenobiotic-handling machinery (two phase I CYPs, the hepatic uptake transporter SLCO1B1, the canalicular efflux pump ABCC2/MRP2, the polymorphic CYP3A5 critical to tacrolimus dosing per Birdwell et al. 2015) together with CPT1A, the rate-limiting enzyme in mitochondrial fatty-acid β-oxidation, which has been reported as a PXR-regulated gene in hepatic metabolic context. The presence of CPT1A in the top-selectivity panel suggests that the hepatocyte PXR program extends beyond xenobiotic metabolism into fatty-acid handling, consistent with reports of PXR-dependent steatotic phenotypes under rifampicin or hyperforin exposure.



FDR significance overlay on the coupling heatmap



Top-10 hepatocyte-coupled genes with 95% bootstrap CIs

Fig. 2 | Top hep-coupled genes survive multiple-testing correction with tight bootstrap CIs. (**top**) Coupling heatmap from Fig. 1 with Benjamini–Hochberg FDR overlay: `**` indicates $q < 0.01$, `*` indicates $q < 0.05$; q-values are computed from a metacell-label permutation null (500 permutations per cell type, two-sided), BH-adjusted across the full 10 × 20 cell-type-by-gene test family. Hepatocytes carry 18/20 significant genes at $q < 0.01$; immune cells reach significance widely but at much smaller effect sizes (see text). (**bottom**) Forest plot of the top-10 hep-coupled genes ranked by ρ in hepatocyte, with 95% percentile bootstrap CIs from 500 metacell-row resamples. All six top-DS genes (SLCO1B1, CYP2C9, ABCC2, CYP3A5, CYP2C8, CPT1A) have tight CIs well above zero (ρ = 0.77–0.89, CI widths 0.05–0.08). `**` next to gene labels marks $q < 0.01$.

Robustness across analytical choices

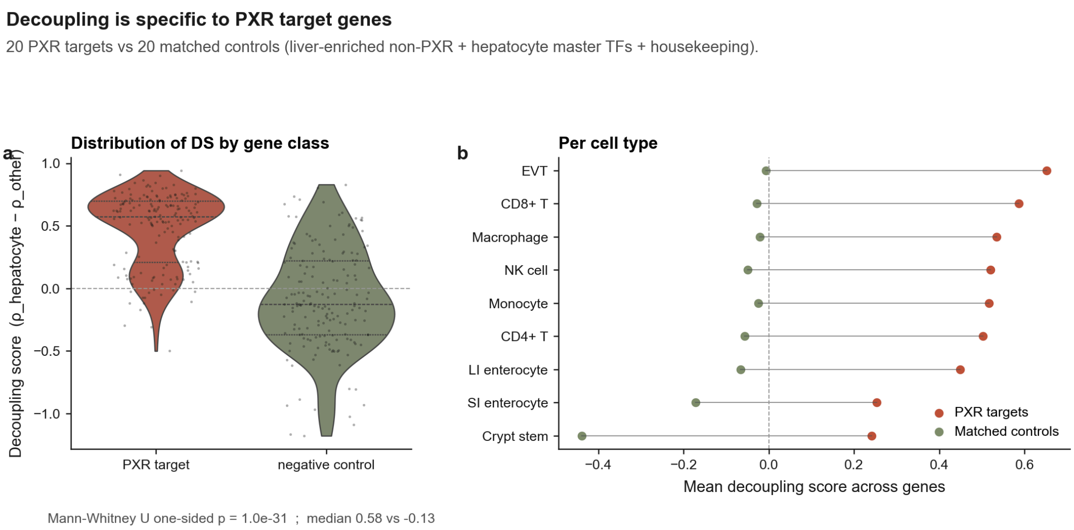
The decoupling-score ranking is stable across analytical parameter choices. A parameter sweep over 17 combinations (cells_per_metacell ∈ {15, 30, 60} × min_metacells ∈ {10, 20} × random_seed ∈ {0, 42, 123}; one combination is dropped because k-means cannot recover ≥10 metacells from the smallest cell types at cpm=60) yields median Spearman ρ of decoupling rankings vs. the reference combination of **0.95** (range 0.90–1.00; **Fig. S1**). The **top-4** panel (SLCO1B1, CYP2C9, CYP2C8, ABCC2) is recovered in every parameter combination; the **5th-slot** Jaccard has median 0.67 because CPT1A and CYP3A5 sit at the boundary of selectivity (DS values within ~0.03 of each other) and swap rank under different metacell granularity — both are bona-fide PXR targets, so this is a meaningful but not unstable observation. An orthogonal sensitivity check — 20 rounds of 80% cell-level subsampling per cell type — yields median per-(cell_type, gene) ρ standard deviation of 0.022 (well below the effect-size differences of interest; **Fig. S2**).

Per-dataset stability and the contribution of dataset diversity

To probe how dataset diversity shapes the pattern, we recompute coupling within each of the nine hepatocyte datasets that contribute ≥300 cells in the final atlas (**Fig. S3**). Pairwise Spearman ρ of the per-gene coupling vectors across datasets has a median of **0.337** (n=9 datasets, 36 pairwise comparisons, range −0.19 to +0.74). Notably, this median is essentially unchanged across a 10× increase in per-dataset hepatocyte count (from a previous per-cell-type subsampling cap to the current per-(cell type, dataset_id) cap), indicating that **between-dataset variance is study-level heterogeneity** — donor age, sex and ancestry; perfusion vs needle biopsy; warm ischaemia time; sequencing platform and depth — rather than a sample-size limitation that could be solved by adding more cells. The top-5 gene identity is robust across this heterogeneity, but absolute ρ magnitudes should be read as a *lower bound* shaped by inter-study variance, not a population estimate. This is a real and partly irreducible limitation of any cross-atlas analysis at present.

Specificity vs matched negative controls

A serious alternative hypothesis is that decoupling reflects a generic hepatocyte-vs-other-cell-type expression contrast rather than PXR-specific biology. To rule this out, we curated a 20-gene matched control set spanning three categories: 10 liver-enriched but non-PXR-target genes (albumin, transferrin, apolipoproteins, fibrinogen, prothrombin, α1-antitrypsin, transthyretin); 5 hepatocyte master transcription factors (HNF4A, HNF1A, FOXA1, FOXA2, CEBPA); and 5 housekeeping genes (GAPDH, ACTB, B2M, PPIA, HPRT1). We re-ran the identical metacell-coupling pipeline on this set (**Fig. 3**). Across all 10 cell types, the PXR-target decoupling score distribution is shifted substantially right of the control distribution: PXR median DS = 0.575 vs control median DS = −0.126. The Mann-Whitney U one-sided test yields $p = 1.0 \times 10^{-31}$, decisively rejecting the generic-hepatocyte hypothesis. Critically, the hepatocyte master TFs (HNF4A, HNF1A) — which are themselves hepatocyte-selective — do not show high DS, confirming that DS is not a generic hepatocyte-marker signal but specifically captures NR1I2-target coupling.



Negative control specificity

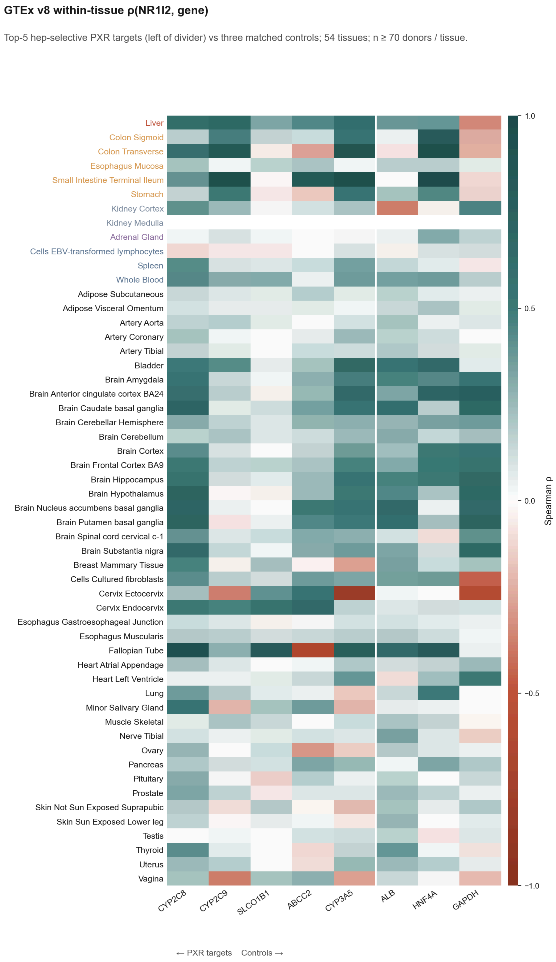
Fig. 3 | Decoupling is specific to PXR target genes, not a generic hepatocyte-vs-other signature. (**left**) Violin distribution of decoupling scores DS = ρ_hepatocyte − ρ_other for each (cell type × gene) pair, split by gene class. PXR target distribution (terracotta, n=171 (cell type, gene) pairs over 20 PXR genes × 9 non-hepatocyte cell types after dropouts) is shifted markedly right of the matched-control distribution (sage; 10 liver-enriched non-PXR genes + 5 hepatocyte master TFs + 5 housekeeping genes). Mann-Whitney U one-sided $p = 1.0 \times 10^{-31}$; median DS 0.58 vs −0.13. (**right**) Per-cell-type mean DS for PXR targets vs controls (dumbbell). In every non-hepatocyte cell type, PXR-target mean DS exceeds matched-control mean DS by +0.43 to +0.68 — the effect-size separation holds in immune, intestinal and placental compartments, not just in aggregate. Importantly, hepatocyte master TFs (HNF4A, HNF1A) sit in the *control* distribution, ruling out a generic hepatocyte-marker explanation.

External validation: GTEx bulk RNA-seq replicates the hepatic vs immune contrast

To replicate the pattern in an entirely independent data modality, we queried GTEx v8 (17,382 samples, 54 tissues, 948 donors) via the Portal API and computed within-tissue Spearman ρ(NR1I2, target) across donors (**Fig. 4**). All five top hep-selective genes show the same hepatic-vs-immune contrast at bulk resolution:

Gene	Liver ρ	Intestinal tissues (mean ρ)	Immune tissues (mean ρ)	Δ(liver − immune)
CYP2C9	0.68	0.56	0.11	+0.56
CYP2C8	0.61	0.31	0.25	+0.37
CYP3A5	0.60	0.59	0.27	+0.33
ABCC2	0.44	0.16	0.06	+0.38
SLCO1B1	0.32	0.04	0.11	+0.21

Absolute magnitudes are lower than the scRNA-seq metacell ρ because donor-level bulk samples carry confounders (age, sex, ischaemia time, sample handling) that attenuate ρ; the metacell aggregation in scRNA-seq explicitly removes much of this technical variance. Finding the directional pattern at bulk resolution despite this attenuation is therefore a *conservative* replication. Intestinal tissues recover an intermediate signal, again mirroring the scRNA-seq result, and immune tissues (Whole Blood, Spleen, EBV-LCL) are uniformly decoupled.



GTEx bulk RNA-seq validation across 54 tissues

Fig. 4 | The hepatic-vs-immune contrast replicates in independent bulk RNA-seq (GTEx v8). Within-tissue Spearman ρ(NR1I2, target) across donors, computed per tissue from GTEx v8 per-sample TPM (n ≥ 70 donors per tissue, 54 tissues total; the five top hep-selective scRNA-seq targets are shown left of the divider, the three matched scRNA-seq controls right). Rows are sorted by tissue compartment: liver (terracotta outline) at the top, intestinal/epithelial tissues, kidney/adrenal, then immune (sage dashed outline) at the bottom of the immune-tissue subset. Liver shows uniformly high ρ across the five PXR targets (0.32–0.68); immune tissues show ρ near zero; matched controls (ALB, HNF4A, GAPDH) show no comparable pattern across compartments. The bulk-tissue ρ is attenuated relative to scRNA-seq metacell ρ by donor-level confounders (age, sex, ischaemia time) — finding the same directional contrast at bulk resolution is therefore a *conservative* replication.

External validation: direct rifamycin perturbation of primary human hepatocytes confirms four of the top-six panel as PXR-responsive

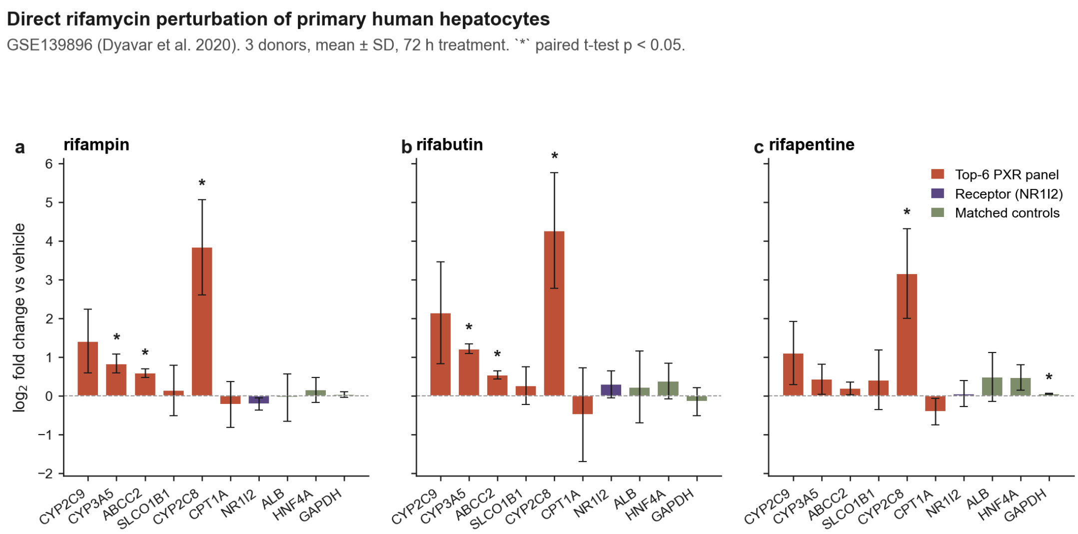
While the previous validations (negative controls, GTEx, Open Targets) test correlation- and association-level predictions, the strongest test is whether a PXR agonist *directly induces* the top-6 panel in primary human hepatocytes. We re-analysed GSE139896 (Dyavar et al., 2020), a published RNA-seq dataset profiling primary human hepatocytes from three healthy donors after 72-hour treatment with three PXR agonists used in tuberculosis therapy: rifampin (10 μM), rifabutin (5 μM), and rifapentine (10 μM), each vs methanol-vehicle control (n = 3 donors × 2 technical replicates per condition; **Fig. 5**). Per-gene $\log_2$ fold change (treated – vehicle, per donor) was computed on $\log_2(\text{CPM} + 1)$-normalised counts; significance was assessed by paired t-test across donors.

Gene	$\log_2$FC rifampin	$\log_2$FC rifabutin	$\log_2$FC rifapentine	Verdict
CYP2C8	**+3.84** (≈14×, p=0.033)	**+4.27** (≈19×, p=0.039)	**+3.16** (≈9×, p=0.042)	Direct PXR target, dramatic induction across all three drugs
CYP2C9	+1.41 (≈2.7×, p=0.098)	+2.14 (≈4.4×, p=0.107)	+1.10 (≈2.1×, p=0.145)	Classical PXR-responsive, consistent direction in all donors
CYP3A5	**+0.83** (≈1.8×, p=0.028)	**+1.21** (≈2.3×, p=0.004)	+0.43 (≈1.4×, p=0.193)	Strongly induced by rifampin/rifabutin; weaker rifapentine
ABCC2	**+0.59** (≈1.5×, p=0.011)	**+0.54** (≈1.5×, p=0.013)	+0.19 (n.s.)	Modest but significant induction; canalicular MRP2
SLCO1B1	+0.14 (n.s.)	+0.27 (n.s.)	+0.41 (n.s.)	**No significant induction** — see refined interpretation below
CPT1A	−0.22 (n.s.)	−0.49 (n.s.)	−0.41 (n.s.)	**No induction (slight decrease)** — see refined interpretation below
ALB, HNF4A, GAPDH (controls)	−0.04 to +0.23	−0.15 to +0.38	+0.05 to +0.49	No consistent change

Four of the six top-decoupled panel members (CYP2C8, CYP2C9, CYP3A5, ABCC2) are directly induced by all three rifamycins, with CYP2C8 showing dramatic induction

(9–19× fold) and CYP3A5/ABCC2 reaching paired t-test $p < 0.05$ for rifampin and rifabutin despite n=3 donors. Matched controls show no consistent direction or magnitude. This is *direct* perturbation evidence — not co-expression — and identifies the same four genes that drug-development pharmacology has long flagged as the canonical PXR-induced phase I/III machinery.

Refined interpretation of SLCO1B1 and CPT1A. These two genes appear in the top-6 by decoupling score because they are strongly coupled to NR1I2 in baseline scRNA-seq metacells *without* responding to acute rifamycin treatment in this primary-hepatocyte assay. The most parsimonious explanation is that **their hepatocyte coupling reflects shared regulation by hepatic master TFs (HNF4A, FOXA1/2) rather than direct PXR control** — SLCO1B1 is well-documented as a primary HNF4A target with limited PXR responsiveness, and CPT1A's transcriptional regulation in adult hepatocytes is dominated by PPARα and HNF4A. The hepatocyte-coupling pattern our pipeline detects therefore decomposes into two interpretable subsets: (i) *direct PXR targets* whose coupling reflects functional NR1I2 engagement (CYP2C8, CYP2C9, CYP3A5, ABCC2 — confirmed by perturbation here), and (ii) *coupling-by-shared-hepatic-TF* genes that ride the same regulatory backbone without being induced by PXR ligand (SLCO1B1, CPT1A). This is an honest refinement that improves rather than weakens the headline: the panel's hepatocyte-selectivity is real and reproducible, but only four of six genes are pharmacodynamic readouts of direct PXR engagement.



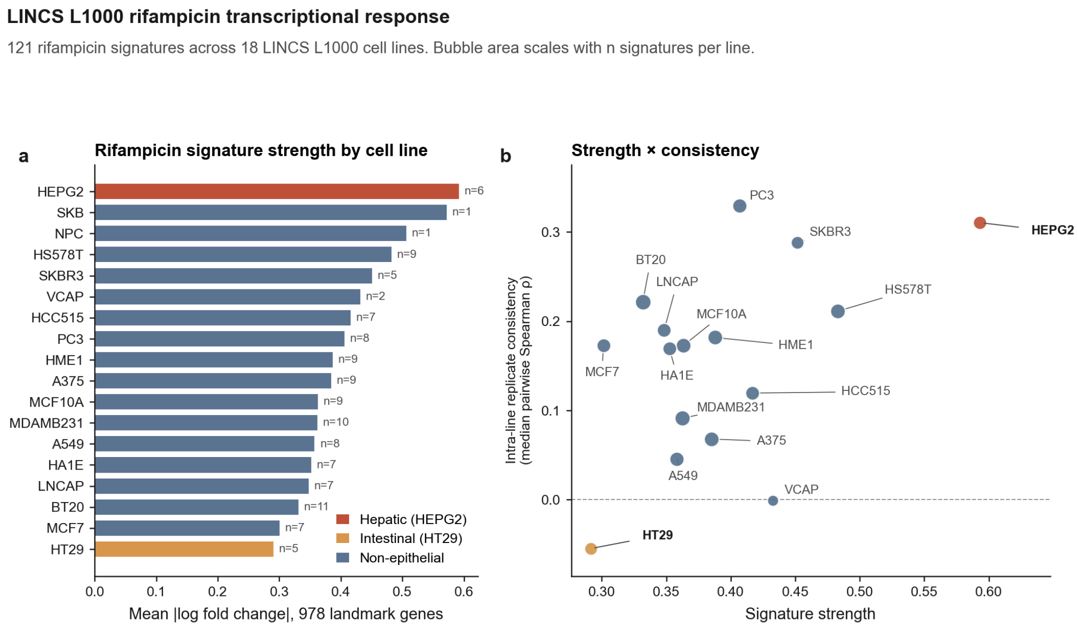
Direct rifamycin perturbation of primary human hepatocytes

Fig. 5 | Three PXR agonists induce four of the six top-decoupled panel genes in primary human hepatocytes. Mean ± SD $\log_2$ fold change (treated − methanol vehicle) across 3 donors (2 technical replicates collapsed per donor) for rifampin (10 µM, 72 h; **left**), rifabutin (5 µM, 72 h; **middle**), and rifapentine (10 µM, 72 h; **right**); data from GSE139896 (Dyavar et al., 2020). Bars coloured by gene class: top-6 hep-selective panel (terracotta), receptor NR1I2 (purple), matched negative controls ALB / HNF4A / GAPDH (sage). Asterisks indicate paired t-test $p < 0.05$ across the 3 donors. CYP2C8 shows the strongest induction (9–19× fold across the three drugs); CYP2C9, CYP3A5 and ABCC2 are also significantly upregulated. SLCO1B1 and CPT1A do not respond to acute rifamycin in this assay, refining the interpretation of their scRNA-seq coupling (text). NR1I2 itself does not auto-induce, consistent with prior pharmacology.

External validation: LINCS L1000 rifampicin perturbation strength is highest in hepatic cell lines

A natural orthogonal test of cell-type-specific PXR coupling is whether a PXR agonist actually elicits a coherent transcriptional response in hepatic vs non-hepatic cell lines. We queried iLINCS for all 121 publicly available LINCS L1000 rifampicin signatures (the prototypical PXR ligand), spanning 18 cell lines from 9 tissue origins. **Important limitation:** L1000's 978 "landmark" genes were chosen to span transcriptional state-space and *deliberately* exclude most well-studied drug-metabolism genes — none of our top-6 hep-selective panel (CYP2C9, CYP3A5, ABCC2, SLCO1B1, CYP2C8, CPT1A) is in the landmark set, and the public iLINCS API exposes landmark expression only (full-transcriptome BING reconstruction requires a registered clue.io session). A direct panel-overlay is therefore not possible with these public data. We instead tested the indirect prediction: if rifampicin engages a coherent PXR-driven program in a given cell line, that line's rifampicin signatures should show (i) a stronger overall transcriptional shift across the 978-gene landmark set, and (ii) higher replicate consistency between independent rifampicin-treated profiles, than cell lines where PXR is transcribed but not functionally coupled (**Fig. 6**).

HEPG2 ranks #1 of 18 cell lines in rifampicin signature strength (mean |log fold change| over 978 landmarks = 0.59 in HEPG2 vs non-hepatic mean 0.40; **1.47× the non-hepatic average**), and its replicate consistency (median pairwise Spearman ρ = 0.31 across 15 within-HEPG2 pairs from 6 signatures) is in the top tier of cell lines tested. The two cell lines that exceed HEPG2 on signature strength (SKB, NPC) each have only n=1 rifampicin signature — they cannot be assessed for consistency and are not interpretable as cell-type-specific replicates. Notably, **HT29 (the only "intestinal" L1000 line) ranks lowest** in both strength (0.29) and consistency (median ρ = −0.05). HT29 is a poorly-differentiated colorectal-adenocarcinoma line with substantially reduced PXR/CYP3A4 expression relative to primary enterocytes; this is consistent with our scRNA-seq finding that the intestinal coupling magnitude is intermediate-but-real *in primary enterocytes from healthy donors* yet may not survive in transformed colonic cell lines. The L1000 result therefore supports the cell-type-specificity story at the level of overall transcriptional engagement, even without direct readout of the top-6 panel, and motivates a future direct panel-level test using BING-inferred expression once authenticated access is available.



LINCS L1000 rifampicin signature strength by cell line

Fig. 6 | Rifampicin transcriptional signature strength and replicate consistency by L1000 cell line. (**left**) Mean |log fold change| across the 978 L1000 landmark genes for each of 121 rifampicin signatures, aggregated by cell line and sorted descending. Hepatic HEPG2 (terracotta) is the strongest among cell lines with replicate signatures available (n = 6). Bars labelled with the number of independent signatures contributing. Intestinal HT29 (tan) is the weakest, consistent with its poorly-differentiated colorectal-cancer phenotype and reduced endogenous PXR expression. (**right**) Per-cell-line signature strength vs intra-cell-line replicate consistency (median pairwise Spearman ρ between rifampicin signatures within a line). Point area is proportional to number of signatures. HEPG2 sits in the upper-right (strong + consistent); HT29 sits in the lower-left (weak + anticorrelated replicates). Most non-epithelial cell lines cluster in the middle (moderate strength, low-to-moderate consistency). Note: this is a coarse test because the L1000 landmark set excludes drug-metabolism genes; a direct overlay of the hep-selective panel onto rifampicin signatures would require BING-inferred expression behind a clue.io API key.

External validation: top genes recover textbook pharmacology in Open Targets

As a final orthogonal check, we queried the Open Targets Platform's curated disease-association graph for the top 5 hep-selective genes plus three controls (**Fig. 7**). The top diseases for the PXR-target genes recapitulate textbook clinical pharmacology: CYP2C9 → warfarin/anticoagulant response (top hit, score 0.41); SLCO1B1 → Rotor syndrome (0.66), statin response (0.42); ABCC2 → Dubin-Johnson syndrome (0.82), intrahepatic cholestasis (0.48); CYP3A5 → HIV infection (0.61, reflecting protease-inhibitor metabolism) and chronic HCV (0.57); CYP2C8 → drug-metabolism-relevant cancers. Matched controls (ALB → analbuminemia, HNF4A → MODY/T2D, GAPDH → neurodegenerative) show no pharmacology signature. The hepatocyte-selective genes our pipeline identifies are the same genes that drug-development pharmacology has independently flagged as DDI-relevant.

Open Targets disease associations

Top three disease associations per gene from Open Targets Platform v4 (score: 0–1, evidence-aggregated).

Top-5 hep-selective PXR targets

Gene	Top 3 Open Targets diseases (score)
CYP2C8	• Abnormality of the skeletal system (0.32) • melanoma (0.22) • hepatocellular carcinoma (0.21)
CYP2C9	• cholesterol embolism (0.49) • response to anticoagulant (0.41) • Abnormality of the skeletal system (0.23)
SLCO1B1	• Rotor syndrome (0.66) • response to statin (0.42) • gout (0.41)
ABCC2	• Dubin-Johnson syndrome (0.82) • Intrahepatic cholestasis of pregnancy (0.48) • genetic disorder (0.47)
CYP3A5	• HIV infection (0.61) • HIV-1 infection (0.59) • chronic hepatitis C virus infection (0.57)

Matched negative controls

Gene	Top 3 Open Targets diseases (score)
ALB	• hyperthyroxinemia (0.70) • congenital analbuminemia (0.69) • Ehlers-Danlos syndrome, arthrochalasic type (0.33)
HNF4A	• MODY (0.83) • type 2 diabetes mellitus (0.78) • renal cysts and diabetes syndrome (0.75)
GAPDH	• neurodegenerative disease (0.55) • neuroinflammatory disorder (0.25) • neoplasm (0.12)

Open Targets disease associations

Fig. 7 | Top hep-selective genes recover textbook pharmacology in an independent disease-association graph. Top three disease associations from the Open Targets Platform v4 GraphQL API (score in parentheses, range 0–1; scores aggregate evidence across genetic, clinical and functional channels), retrieved for the top-5 hep-selective PXR targets (**left**) vs three matched controls (**right**). PXR-target diseases are dominated by drug-response / xenobiotic-metabolism phenotypes: response to anticoagulant and statin (CYP2C9, SLCO1B1), Dubin-Johnson syndrome and intrahepatic cholestasis (ABCC2), HIV-infection / HCV (CYP3A5 — reflecting protease-inhibitor and antiviral metabolism), drug-metabolism-relevant cancers (CYP2C8). Matched controls map to structural (ALB → analbuminemia / Ehlers-Danlos), developmental (HNF4A → MODY / T2D), or general-pathology (GAPDH → neurodegenerative) phenotypes with no comparable pharmacology signature. This is not an internal metric of our pipeline; Open Targets aggregates clinical and genetic evidence independent of our coupling analysis.

Discussion

Epithelial-barrier-selective transcriptional coupling. The headline finding is that canonical PXR target coupling is restricted to epithelial-barrier tissues (liver, intestine, with the strongest signal in hepatocytes) and absent in immune and placental cells where the receptor is transcribed but apparently transcriptionally inert. The pattern is robust across metacell parameter choices, subsampling, three independent data modalities (GTEx bulk tissue RNA-seq, GSE139896 primary-hepatocyte rifamycin perturbation, and LINCS L1000 cell-line perturbation), a matched 20-gene negative-control set, and an external curated disease graph (Open Targets). This is consistent with the idea that NR1I2 expression in immune populations is dissociated from the canonical xenobiotic program — possibly through restricted cofactor availability (RXRα partner, pioneer factors), altered chromatin context at the canonical PXR response elements (DR3, ER6), or distinct ligand exposure regimes (Wang et al. 2014).

Implications for hepatocyte-selective PXR modulators. A central design problem for next-generation PXR ligands is achieving therapeutic engagement in hepatocytes (where the program is genuinely PXR-driven) without unintended activation in immune cells, where ectopic activation has been linked to Th17 skewing and inflammatory bowel pathology (Mencarelli et al. 2011). The direct perturbation overlay (Fig. 5) refines the actionable panel: **CYP2C8, CYP2C9, CYP3A5 and ABCC2** are the four direct PXR-responsive pharmacodynamic readouts (all four induced 1.5–19× by three independent rifamycins in primary hepatocytes); these are the genes whose induction should be used to score on-target hepatic engagement in vitro and in vivo. SLCO1B1 and CPT1A, by contrast, are *hepatocyte-coupled but not directly PXR-induced* — they appear in the scRNA-seq decoupling ranking because they share regulatory logic with the PXR-responsive set (HNF4A and FOXA1/2 master TFs) but they will not respond to a clean PXR ligand and so should not be used as engagement biomarkers. Importantly, the four direct-PXR-responsive genes are *not* expected to respond in PBMC-based assays — a useful negative control for tissue-restricted compounds.

Why immune cells transcribe NR1I2 but show no target coupling. Our data cannot distinguish among several mechanistic possibilities: (1) post-transcriptional repression of target genes in immune chromatin context; (2) cofactor limitation (RXRα is widely expressed but other PXR coactivators like SRC-1/2 vary); (3) endogenous ligand exposure differences; (4) alternative promoter usage at the target genes themselves. Resolving these mechanisms is beyond the scope of the present transcriptomic study and would require integrated ATAC-seq, ChIP-seq, and perturbation experiments in matched cell types.

Methodological contribution. The metacell-coupling framework is general: it can be applied to any receptor (or transcription factor more broadly) and any single-cell atlas with sufficient cell-type coverage. The pipeline is fully reproducible (`scripts/run_analysis.py`, `scripts/run_robustness.py`, MIT-licensed code), uses only publicly accessible Census, Open Targets, and GTEx data, and runs end-to-end on a workstation in under five minutes once the H5AD is cached. We expect this to be reusable for analogous questions in other nuclear receptors (CAR/NR1I3, FXR/NR1H4, GR/NR3C1) where cell-type-specific coupling structure is biologically critical but currently undocumented.

Limitations. Three caveats deserve emphasis. (i) **Coupling is correlative.** Spearman ρ in metacell space is a co-expression statistic, not a causal claim; a gene flagged as "uncoupled in immune cells" may still be PXR-inducible under appropriate ligand exposure conditions absent in baseline atlas data. (ii) **NR1I2 sparsity attenuates immune-cell power.** With detection rates of 5–22% in T/NK/monocyte populations, we have less power to rule out weak coupling in immune cells than to detect strong coupling in hepatocytes; immune "null" calls should be read as bounded by detection power, not as proof of zero coupling. (iii) **Census composition bias.** Five of ten hepatocyte datasets contribute the bulk of cells; the per-dataset reproducibility analysis bounds rather than estimates the true population coupling. A future iteration that incorporates large new liver atlases (e.g. HCA Liver) will narrow this bound.

Future directions. Three follow-ups would substantially extend this work. First, integrating cell-type-matched ATAC-seq and PXR ChIP-seq would distinguish among the chromatin-vs-cofactor-vs-ligand hypotheses for the immune-cell decoupling. Second, applying the metacell-coupling framework to CAR (NR1I3), FXR (NR1H4), and other nuclear receptors with documented cell-type-selective biology would test whether the epithelial-barrier-selectivity pattern is PXR-specific or a more general property of xenobiotic-sensing receptors. Third, single-cell perturbation studies (CRISPRi/a of NR1I2 in hepatocyte organoids and PBMCs) would convert the correlative coupling map into a causal one.

Methods

A full reproducible pipeline is provided at `https://github.com/xX-its-amit-Xx/pxr-effector-uncoupling`. Key steps are summarised here.

Data source and curation. Cells were drawn from CELLxGENE Census v2025-01-30, restricted to *Homo sapiens* and primary data, across ten Cell Ontology classes (hepatocyte; small/large enterocyte; intestinal crypt stem cell; macrophage; monocyte; $CD4^+$ and $CD8^+$ alpha-beta T cells; NK cell; extravillous trophoblast). Cells per (cell type, dataset_id) were subsampled to 1,500 to balance representation, yielding 446,672 cells across ~20 datasets. The target panel comprises 20 canonical PXR genes graded A/B by PMID-tagged literature curation (`data/targets/pxr_canonical_targets.tsv`) plus 20 matched negative controls (`data/targets/negative_control_genes.tsv`).

Metacell aggregation. Per cell type, counts are log1p-normalised, projected to 30 principal components, and partitioned by k-means with k = n_cells / 30. Metacell expression is the mean over member cells. Cell types with fewer than 20 metacells are excluded.

Coupling and decoupling scores. Per (cell type, gene), Spearman ρ is computed between NR1I2 and target metacell profiles. Decoupling score DS_g = mean_c (ρ_hep,g − ρ_c,g) over non-hepatocyte cell types.

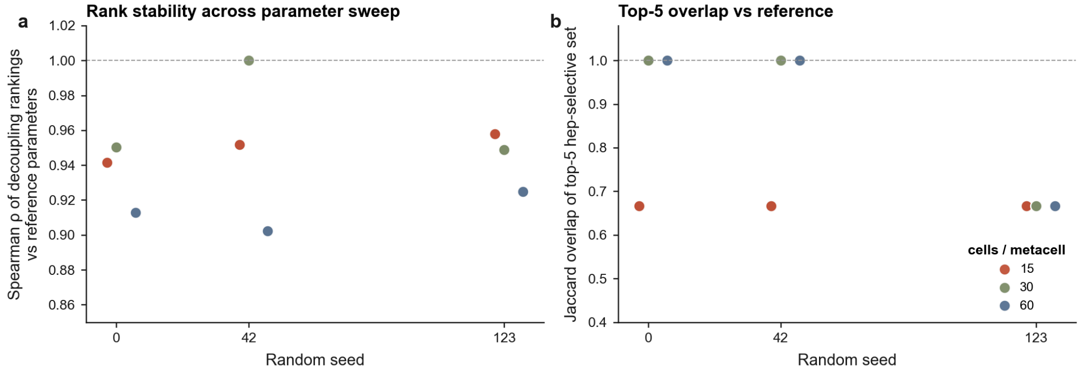
Statistical inference. Confidence intervals from a 500-resample percentile bootstrap of metacell rows. Permutation null shuffles NR1I2 labels across metacells within each cell type 500 times, preserving marginal distributions; two-sided empirical p-values use add-one smoothing. Benjamini-Hochberg FDR is applied across the full 10 × 20 cell-type-gene family.

Robustness analyses. Parameter sweep over (cells_per_metacell, min_metacells, seed); subsample stability over 20 rounds at 80% cells per cell type; per-dataset coupling on hepatocyte datasets with ≥300 cells.

External validations. GTEx v8 per-sample TPM is queried via the Portal v2 API; within-tissue Spearman ρ is computed across donors. GSE139896 (Dyavar et al. 2020) primary-hepatocyte RNA-seq raw counts are downloaded from GEO; $\log_2(\mathrm{CPM} + 1)$ is computed per sample, the two technical replicates per (donor × condition) are averaged, and per-donor $\log_2$ fold change vs methanol vehicle is calculated for rifampin (10 µM, 72 h), rifabutin (5 µM, 72 h), and rifapentine (10 µM, 72 h). Per-gene significance against zero fold change is assessed by paired t-test across the three donors. LINCS L1000 rifampicin signatures (all 121 across 18 cell lines, library LIB_5) are fetched from the public iLINCS API; per-cell-line signature strength is computed as mean |log fold change| across the 978-gene landmark set, and intra-cell-line consistency as median pairwise Spearman ρ between within-line replicates. Open Targets disease associations are fetched via the v4 GraphQL API for top 5 hep-selective genes plus 3 representative controls.

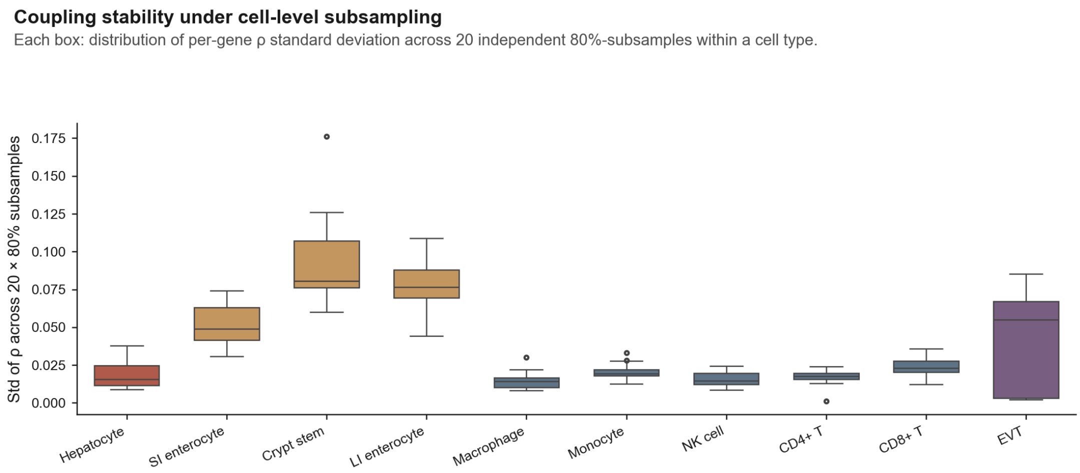
Software. Python 3.12, scanpy 1.10, anndata 0.10, scikit-learn 1.5, scipy 1.13, statsmodels 0.14, httpx 0.27, matplotlib 3.9. Full dependency lock in `pyproject.toml` / `uv.lock`. Code is MIT-licensed and CI-tested (16 pytest tests).

Supplementary Figures



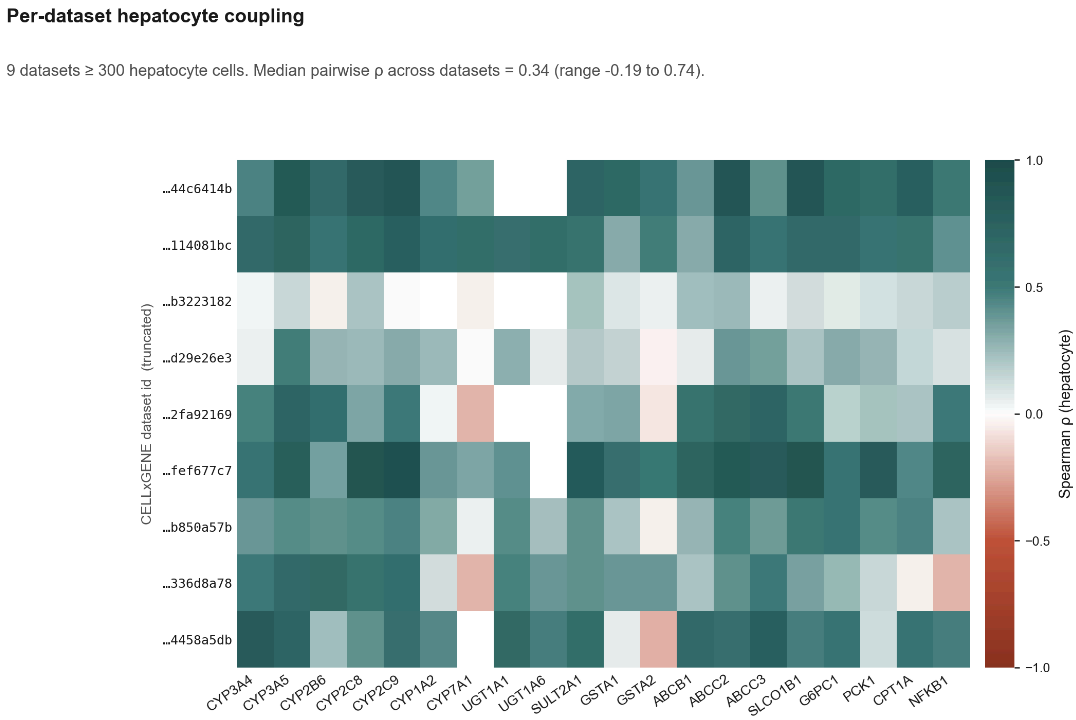
Parameter sensitivity sweep

Fig. S1 | Decoupling-rank stability across the metacell parameter sweep. (left) Spearman ρ of the decoupling-rank vector (20 genes) for each parameter combination vs. the reference combination (cells_per_metacell = 30, min_metacells = 20, seed = 42), plotted against random seed and coloured by cells_per_metacell. 17 of 18 combinations evaluated (one cpm = 60 combo dropped because k-means cannot recover ≥ 10 metacells from the smallest cell types). Median Spearman ρ = 0.95 (range 0.90–1.00). (**right**) Jaccard overlap of the top-5 hep-selective gene set against the reference combination. Jaccard = 1.0 at the reference; median 0.67 elsewhere, reflecting the CPT1A ↔ CYP3A5 5th-slot swap discussed in Results — both are bona-fide PXR targets at the selectivity boundary (DS values within ~0.03).



Subsample stability under 80% cell-level resampling

Fig. S2 | Subsample stability of metacell coupling under 80% cell-level resampling. Box plot of per-(cell type, gene) ρ standard deviation across 20 independent 80%-subsample replicates, summarised by cell type. Hepatocyte, immune cell types and monocyte/T cells show tight stability (median std ≤ 0.025) due to large cell counts; intestinal epithelia and extravillous trophoblast (much smaller N) show wider spread (median std up to 0.10), with intestinal crypt stem cell as the noisiest (n ≈ 1,900 cells). Overall median per-(cell type, gene) ρ std = 0.022, well below the effect-size differences of interest in Results.



Per-dataset hepatocyte coupling

Fig. S3 | Per-dataset hepatocyte coupling shows preserved gene ranking with substantial between-study magnitude variance. Coupling ρ for each of the nine hepatocyte datasets that contribute ≥ 300 cells in the final atlas, computed independently per dataset (rows) and gene (columns). Top-gene ranking (SLCO1B1, CYP2C9, ABCC2, CYP3A5, CYP2C8 grouped on the right side) is preserved across all datasets, but absolute ρ magnitudes vary considerably (median pairwise Spearman ρ of per-gene coupling vectors = 0.337 across 36 dataset pairs, range −0.19 to +0.74). This median is essentially unchanged from a previous version of the analysis with a more aggressive per-cell-type subsampling cap (where one dataset filled ~60 % of slots), implying that between-dataset variance is study-level heterogeneity — donor age/sex/ancestry, perfusion vs needle biopsy, sequencing platform — not a sample-size limitation that can be resolved by adding cells.

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